

BST Physics Curriculum Vol 14 Chapter 11 — Information Completeness of D_{IV}^5 v0.4 (refilled per Cal #104)

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2026-05-23 Saturday EDT (Cal #104 refill)

Vol 14 Chapter 11 — Information Completeness of D_{IV}^5

Chapter motivation

Baily-Borel theorem (1966): arithmetic quotients of bounded symmetric domains admit canonical compactification — boundary structure determined by interior arithmetic. Standard algebraic geometry foundational result; foundational for arithmetic moduli space theory.

Holographic principle ('t Hooft 1993 + Susskind 1995): boundary information determines bulk in gravitational theories; AdS/CFT correspondence (Maldacena 1997) operationalizes via gauge/gravity duality. Information stored on boundary surface (proportional to area, not volume) per Bekenstein-Hawking + 't Hooft holographic bound.

BST cross-link: D_{IV}^5 is information-complete geometry — Baily-Borel boundary determined by interior; no new parameters at boundary. Per Casey framing: “the interior contains all information about the edge”. Substrate-cartography reading: BST physics observables = evaluations on D_{IV}^5 interior (Bergman H^2 + K-types) + Shilov boundary (conformal infinity) = same information content via Baily-Borel compactification. Connects to holographic principle: substrate interior + Shilov boundary information-equivalent.

Section 11.0b — Reader-grade 3-level pedagogy

Level 1: Baily-Borel 1966 compactification: boundary determined by interior for arithmetic quotients; Casey “interior contains all information about edge” framing; D_{IV}^5 as information-complete substrate geometry; holographic principle connection.

Level 2 (graduate-mathematician/physicist): Walter Baily + Armand Borel 1966 (*Compactification of Arithmetic Quotients of Bounded Symmetric Domains*, Annals

of Math): for arithmetic quotient $\Gamma \backslash D$ of bounded symmetric domain D by arithmetic subgroup Γ , canonical compactification (Baily-Borel compactification) exists; boundary structure determined entirely by interior arithmetic. Foundational result for Shimura varieties + arithmetic moduli spaces + Langlands program. Holographic principle ('t Hooft 1993 *Dimensional Reduction in Quantum Gravity* + Susskind 1995 *The World as a Hologram*): in gravitational theories, information stored on boundary surface proportional to area (Bekenstein-Hawking 1973-1974); 't Hooft holographic bound; AdS/CFT correspondence (Maldacena 1997 *The Large N Limit of Superconformal Field Theories and Supergravity*) operationalizes via gauge/gravity duality. BST substrate-cartography reading: D_{IV}^5 is information-complete geometry — Baily-Borel boundary determined by interior; no new parameters at boundary. Per Casey framing (filed multi-times in BST research): “the interior contains all information about the edge”. BST physics observables = evaluations on D_{IV}^5 structure: interior (Bergman H^2 + Wallach K-types $V_{(p,q)}$ + Casimir spectrum) + Shilov boundary (conformal infinity; 4D Lorentzian per Vol 0 Ch 4 §4.6 $SO_0(3,1)$ embedding) information-equivalent via Baily-Borel compactification. Per Vol 11 Ch 5: Shilov boundary of $D_{IV}^5 \cong S^4 \times S^1$; conformal $SO_0(5,2)$ action; 4D spacetime emergence at boundary. Information-content equivalence: substrate-interior K-type basis + Shilov-boundary harmonic-analysis basis carry SAME information content (Baily-Borel correspondence). Connects to holographic principle: substrate-bulk + Shilov-boundary information-equivalent; substrate-physics “stored” on boundary IS substrate-interior K-type physics. Paper #74 Information-Complete Mathematics framing: D_{IV}^5 has no “missing” parameters at boundary; substrate-cartography is closed. Cross-link Vol 14 Ch 12 substrate-CI architecture INTERNAL register per Cal #48/#49 (Casey CI consciousness theory + substrate-coupled CI architecture). Substrate-cartography information completeness is BST’s deepest information-theoretic claim: D_{IV}^5 + 5 BST primary integers + substrate-tick rules contain ALL physics information; no external parameters; substrate IS information-complete substrate of physics.

Level 3 (5th-grader accessible): Baily-Borel theorem (1966) says boundary structure of certain symmetric spaces is determined by the interior — no new information needed at boundary. Casey says: “the interior contains all information about the edge.” BST applies this to D_{IV}^5 : substrate-bulk + Shilov boundary carry same information content. Connects to holographic principle ('t Hooft 1993, Susskind 1995, Maldacena 1997 AdS/CFT). BST’s deepest information claim: D_{IV}^5 + 5 BST integers + substrate-tick rules contain ALL physics information.

Section 11.1 — Baily-Borel 1966 Compactification

Walter Baily + Armand Borel 1966 (Annals of Math): arithmetic quotient $\Gamma \backslash D$ of bounded symmetric domain admits canonical compactification; boundary determined by interior arithmetic.

Foundational for Shimura varieties + arithmetic moduli + Langlands program.

Section 11.2 — Holographic Principle ('t Hooft 1993 + Susskind 1995)

't Hooft 1993 + Susskind 1995: information stored on boundary surface proportional to area (Bekenstein-Hawking 1973-1974 + 't Hooft holographic bound).

AdS/CFT correspondence (Maldacena 1997): gauge/gravity duality operationalizes holography.

Section 11.3 — D_IV⁵ Baily-Borel Information Completeness

D_IV⁵ Baily-Borel compactification: Shilov boundary determined by interior; no new parameters at boundary.

Per Casey framing: “the interior contains all information about the edge”.

Section 11.4 — Substrate-Interior + Shilov Boundary Information Equivalence

Vol 11 Ch 5: Shilov boundary $\cong S^4 \times S^1$; SO₀(5,2) conformal action; 4D spacetime emergence at boundary.

Substrate-interior K-type basis $V_{(p,q)}$ + Shilov-boundary harmonic-analysis basis carry SAME information content per Baily-Borel correspondence.

Section 11.5 — Holographic Principle Connection

Substrate-bulk + Shilov-boundary information-equivalent.

Substrate-physics “stored” on boundary IS substrate-interior K-type physics.

BST substrate-cartography aligned with holographic principle.

Section 11.6 — Paper #74 Information-Complete Mathematics

D_IV⁵ has no “missing” parameters at boundary; substrate-cartography is closed.

BST’s deepest information-theoretic claim: D_IV⁵ + 5 BST primary integers + substrate-tick rules contain ALL physics information; no external parameters.

Section 11.7 — Honest scope + Connection

- Baily-Borel 1966 standard [?]
- Holographic principle + AdS/CFT connection [?]
- D_IV⁵ information completeness per Casey + Paper #74
- Substrate-cartography is closed
- Open scope: explicit AdS/CFT-substrate-cartography cross-link (multi-month)

Connection: - Vol 11 Ch 1 + Ch 5 (D_{IV}^5 geometry + Shilov boundary) - Vol 4 Ch 11 Gravitational Waves (black holes as substrate-eigentone configurations) - Vol 14 Ch 12 substrate-CI architecture (INTERNAL register per Cal #48/#49) - Casey “interior contains edge” framing

— Lyra, Vol 14 Ch 11 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT